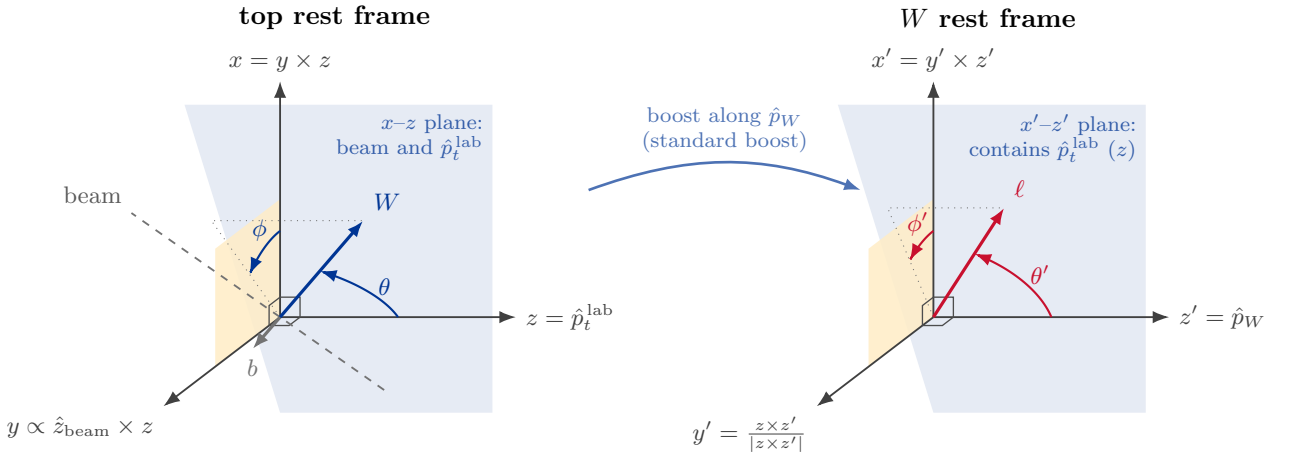


Top rest frame: $z = \hat{p}_t^{\text{lab}}$, $y \propto \hat{z}_{\text{beam}} \times z$ (normal to the plane of the beam and the top), $x = y \times z$; (θ, ϕ) is the direction of the W, ϕ measured from x in the xy plane (gold).
W **rest frame**: $z' = \hat{p}_W$, $y' = z \times z'/|z \times z'|$ lies in the xy plane, $x' = y' \times z'$; (θ', ϕ') is the direction of the charged lepton, i.e. $(x', y', z') = R(\phi, \theta, 0)\,(x, y, z)$.
Hadronic top: the same construction gives $(\tilde{\theta}, \tilde{\phi}, \tilde{\theta}', \tilde{\phi}')$, with the down-type quark in place of ℓ.